LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Equipment setup, tools and spares for Gas welding and Brazing procedure
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Equipment setup, tools and spares for Gas welding and Brazing procedure
5. Specific Objectives	31.1.4.1 Trainee will be able to explain arc welding and brazing procedure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Gas welding and Brazing equipment setup; Safety precautions regarding gas cylinders and torches; Instructor supervision of tool identification and spares verification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between welding and brazing?
(b) List three essential safety precautions for handling gas cylinders.
(c) Identify the two main types of gas welding torches used in aviation maintenance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Gas welding equipment components • Acetylene and oxygen cylinder handling • Regulator and hose assembly setup • Torch tip selection and adjustment • Flame characteristics and control	L & Ds	PPT, IB	15 min
(b)	• Brazing equipment and filler metals • Brazing torch operation and heat control • Preparation of joint surfaces for brazing • Application of flux and filler rod • Cooling and post-brazing inspection	L & Ds	PPT, PDF	20 min
(c)	• Maintenance of welding and brazing tools • Storage and inventory of spare parts • Troubleshooting common equipment faults • Record keeping for tool usage • Final safety check before shutdown	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct sequence for connecting gas cylinders to the welding torch.
(b) Describe the role of flux in the brazing process and why it is necessary.
(c) Identify two common causes of equipment failure during gas welding operations.

12. Home Assignment and Follow Up	Prepare a brief report detailing the specific tools and spares required for a standard aircraft gas welding repair, referencing the next lesson plan on actual welding procedures.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Clarification needed on specific filler metal grades for different aircraft alloys.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min